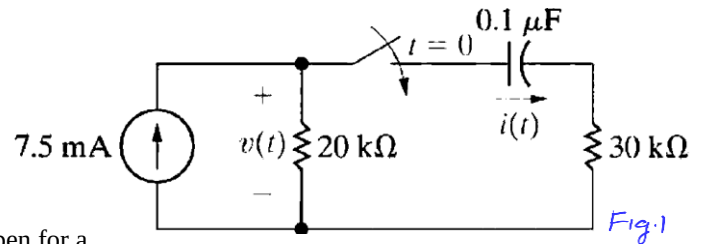
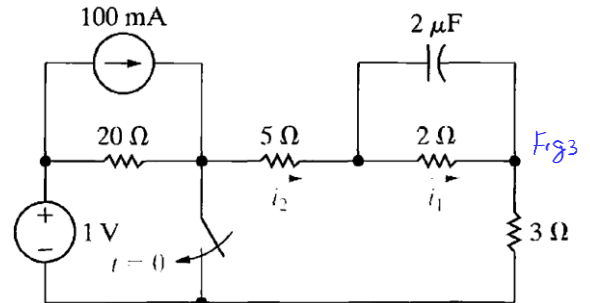
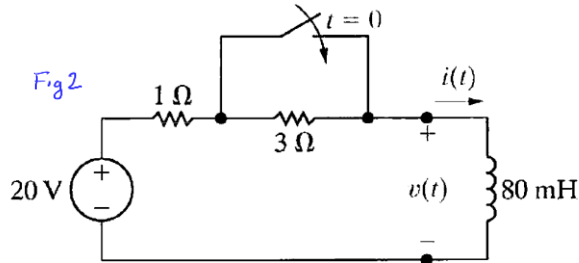


Tutorial-5

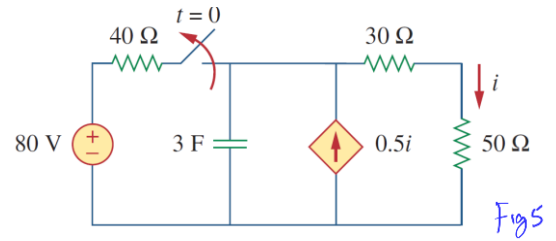
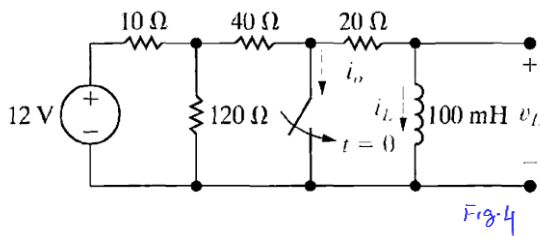
1. The switch in the circuit shown in figure has been open for a long time before closing at $t = 0$. Find $v(t)$ and $i(t)$ for $t \geq 0$



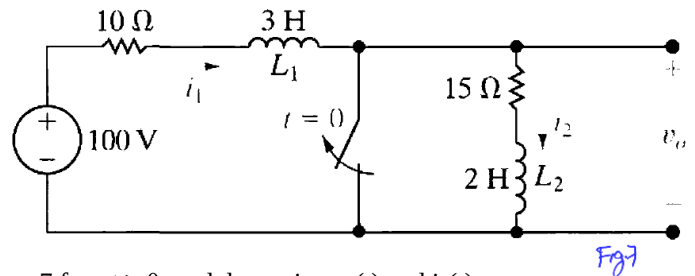
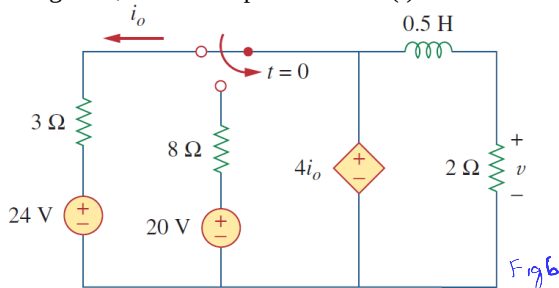
2. The switch in the circuit shown in figure has been open for a long time before closing at $t = 0$. Find $v(t)$ and $i(t)$ for $t \geq 0$



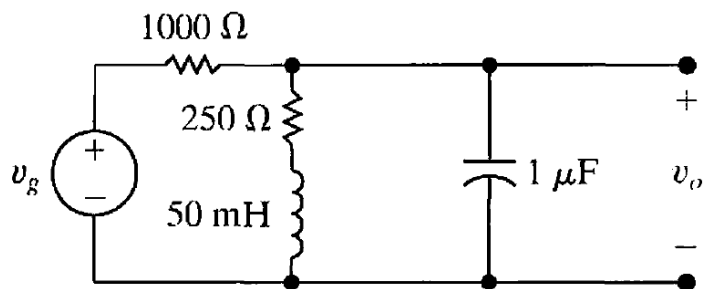
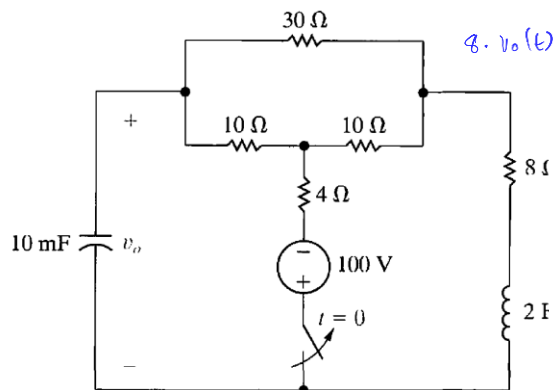
3. The switch in the circuit shown in Figure 3 is closed at $t = 0$ after being open for a long time. Find $i_1(t)$ and $i_2(t)$ for $t \geq 0$
4. The switch in the circuit shown (Fig 4) is closed at $t = 0$ after being open for a long time. Find $i_L(t)$ and $v_L(t)$



5. In Figure 5, find the expression for $i(t)$ for $t \geq 0$
6. In Figure 6, find the expression for $v(t)$ for $t \geq 0$



7. Draw an s-domain circuit for the circuit shown in Figure 7 for $t \geq 0$ and determine $v_0(t)$ and $i_1(t)$



9. Find the tf of the circuit and obtain the poles, zeros, damping coeff and natural frequency.

$$1. v_c(t) = 150 - 150e^{-200t} \text{ V}, \quad i = 3e^{-200t} \text{ mA} \quad v(t) = 150 - 60e^{-200t} \text{ V}$$

$$2. i(t) = 20 - 15e^{-12.5t} \text{ A} \quad v(t) = 15e^{-12.5t} \text{ V}$$

$$3. i_1 = \frac{v_c}{2} = 0.1e^{-312,000t} \text{ A} \quad i_2 = -25e^{-312,000t} \text{ mA}$$

$$5. i(t) = 800e^{-t/480} \text{ mA}$$

$$6. i(t) = 48e^{-4t} \text{ A} \quad v(t) = z i(t) = 96e^{-4t} \text{ V}$$

$$7. v_o(t) = 128(t) + (60 + 10e^{-5t})u(t) \text{ V} \quad i = (4 + 2e^{-5t})u(t) \text{ A}$$

8

$$9. \frac{V_o(s)}{V_g(s)} = \frac{1000(s + 5000)}{s^2 + 6000s + 25 \times 10^6}$$

$$z_1 = -5000 \quad p = -3000 \pm j4000$$